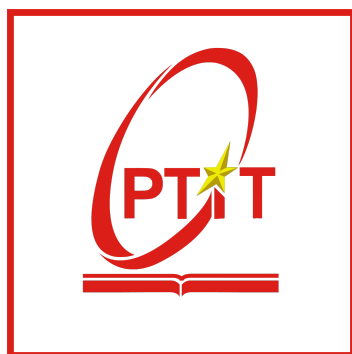


POST AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY 1



Business Intelligence

Project: Business Intelligence System for E-commerce Sales Analysis

Lecture:	Kim Ngọc Bách
Class:	E22HTTT
Group:	03
Members:	Nguyễn Hồng Thăng-B22DCKH118

Hanoi, 2026

Chapter 1: Introduction	3
1.1 Problem Statement	3
1.2 Project Objectives	3
1.3 Scope	3
1.4 Report Structure	3
Chapter 2: Dataset Description	5
2.1 Data Source	5
2.2 Dataset Overview	5
2.3 Entity Relationships	6
2.3.1 olist_orders_dataset.csv	6
2.3.2 olist_order_items_dataset.csv	6
2.3.3 olist_customers_dataset.csv	7
2.3.4 olist_products_dataset.csv	7
2.3.5 olist_sellers_dataset.csv	8
2.3.6 olist_order_reviews_dataset.csv	9
2.3.7 olist_order_payments_dataset.csv	9
2.3.8 product_category_name_translation.csv	10
2.4 Source Table Relationships	10
2.5 Data Quality Observations	11
Chapter 3: Data Warehouse Design	12
3.1 Architecture Overview	12
3.2 Star Schema Design	12
3.2.1 Dimension Tables	12
3.2.2 Fact Table: fact_sales	12
3.3 Reporting Layer	15
Chapter 4: ETL Pipeline	16
4.1 Pipeline Overview	16
4.2 Extract & Load (extract_load.py)	18
4.3 Transform (transform.py)	18
4.4 Build Data Warehouse (build_dw.py)	18
Step 1: Populate Dimension Tables	18
Step 2: Populate Fact Table	18
Step 3: Build Reporting Aggregates	18
4.5 Orchestration with Kestra	19
4.6 Technology Stack	20
Chapter 5: Dashboard and Visualizations	21
5.1 Dashboard Overview	21
5.2 Page 1: Revenue Overview	21
5.3 Page 2: Product Analysis	21
5.4 Page 3: Regional Analysis	22
5.5 Page 4: Operations & Payments	23
Chapter 6: Key Business Insights	25
6.1 Revenue Growth and Seasonality	25
6.2 Product Category Performance	25

6.3 Geographic Distribution	25
6.4 Delivery Performance	25
6.5 Payment Behavior	26
6.6 Seller Performance	26
Chapter 7: Conclusion and Future Work	27
7.1 Summary	27
7.2 Project Achievements	27
7.3 Challenges and Limitations	27
7.4 Future Work	27
References	29
Appendix: Project Structure	30

Chapter 1: Introduction

1.1 Problem Statement

The rapid growth of e-commerce has generated enormous volumes of transactional data that, if properly analyzed, can provide critical business insights. Companies like Olist face challenges in understanding revenue patterns, product performance, regional demand, and operational efficiency across diverse geographic regions.

Traditional reporting methods are insufficient for handling the scale and complexity of modern e-commerce data. A Business Intelligence system is therefore required to automate data collection, transformation, and visualization — enabling data-driven decision making at scale.

1.2 Project Objectives

The primary objectives of this project are:

- Design and implement a Data Warehouse following the Star Schema methodology
- Build an automated ETL pipeline to ingest, clean, and transform raw e-commerce data
- Create aggregate tables and an interactive dashboard to surface business insights
- Analyze key business metrics including revenue trends, product performance, regional distribution, delivery quality, and payment behavior

1.3 Scope

The project is scoped to the Olist Brazilian E-Commerce public dataset available on Kaggle, covering the period from September 2016 to October 2018. The analysis focuses on delivered orders only, providing approximately 96,000 completed transactions for insight generation.

1.4 Report Structure

This report is organized as follows: Chapter 2 describes the dataset; Chapter 3 presents the Data Warehouse design; Chapter 4 details the ETL pipeline; Chapter 5 covers the dashboard and visualizations; Chapter 6 presents key insights; Chapter 7 provides the conclusion and future work.

Chapter 2: Dataset Description

2.1 Data Source

The Olist Brazilian E-Commerce Public Dataset was released by Olist Store under the CC BY-NC-SA 4.0 license (Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International). It is publicly hosted on Kaggle and has been accessed by over 80,000 users worldwide, making it one of the most widely studied e-commerce datasets in academic and data science communities.

Attribute	Detail
Dataset Name	Brazilian E-Commerce Public Dataset by Olist
Publisher	Olist Store (olistbr)
Platform	Kaggle — https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
License	CC BY-NC-SA 4.0 (non-commercial use with attribution required)
Coverage Period	September 2016 — October 2018 (25 months)
Geographic Scope	Brazil — all 27 states and Federal District
Total Files	9 CSV files + 1 auxiliary translation file
Total Records	~1.5 million rows across all files
Download Method	Kaggle API (kaggle datasets download -d olistbr/brazilian-ecommerce)
Last Updated on Kaggle	2018 (static historical snapshot, no live updates)

Olist operates as a marketplace aggregator: small and medium Brazilian retailers list products on the Olist platform, which then publishes listings across major e-commerce platforms (Mercado Livre, Americanas, etc.) under a unified seller profile. This aggregation model means the dataset reflects a diverse range of product categories and geographic seller distributions.

2.2 Dataset Overview

CSV File	Rows	Description
olist_orders_dataset.csv	99,441	Order lifecycle: status, timestamps from purchase to delivery
olist_order_items_dataset.csv	112,650	Line items: product, seller, price, freight per order
olist_customers_dataset.csv	99,441	Customer location: city, state, zip code prefix
olist_products_dataset.csv	32,951	Product attributes: category, dimensions, weight
olist_sellers_dataset.csv	3,095	Seller location: city, state, zip code prefix

CSV File	Rows	Description
olist_order_reviews_dataset.csv	104,719	Customer reviews: score (1-5), comment, timestamps
olist_order_payments_dataset.csv	103,886	Payment details: type, installments, value
olist_geolocation_dataset.csv	~1,000,000	ZIP code to latitude/longitude mapping
product_category_name_translation.csv	71	Portuguese to English category name translation

2.3 Entity Relationships

The following subsections describe each CSV file's columns, data types, and sample values.

2.3.1 olist_orders_dataset.csv

The central table linking all other entities. Each row represents one customer order.

Column Name	Type	Nullable	Description	Sample Value
order_id	VARCHAR(32)	NOT NULL	Unique order identifier (MD5 hash)	e481f51cbdc54678b7cc49136f2d6af7
customer_id	VARCHAR(32)	NOT NULL	FK -> customers.customer_id	9ef432eb6251297304e76186b10a928d
order_status	VARCHAR(50)	NOT NULL	Lifecycle stage of the order	delivered / shipped / canceled
order_purchase_timestamp	TIMESTAMP	NOT NULL	Datetime customer placed the order	2017-10-02 10:56:33
order_approved_at	TIMESTAMP	NULLABLE	Datetime payment was approved	2017-10-02 11:07:15
order_delivered_carrier_date	TIMESTAMP	NULLABLE	Datetime handed to logistics carrier	2017-10-04 19:55:00
order_delivered_customer_date	TIMESTAMP	NULLABLE	Datetime received by customer	2017-10-10 21:25:13
order_estimated_delivery_date	TIMESTAMP	NOT NULL	Estimated delivery date shown to customer	2017-10-18 00:00:00

Status distribution in dataset:

- delivered: ~96,478 orders (97%) — only these are used in analytical pipeline
- shipped: ~1,107 orders (1.1%)
- canceled: ~625 orders (0.6%)

2.3.2 olist_order_items_dataset.csv

Granular line-item data. One order may contain multiple items (up to 21 items in dataset). This table is the primary source for the fact_sales table.

Column Name	Type	Nullable	Description	Sample Value
order_id	VARCHAR(32)	NOT NULL	FK -> orders.order_id	e481f51cbdc54678b7cc49136f2d6af7
order_item_id	INT	NOT NULL	Sequential item number within the order (1-based)	1
product_id	VARCHAR(32)	NOT NULL	FK -> products.product_id	87285cd95629f4e0ed2c553ea24e9483
seller_id	VARCHAR(32)	NOT NULL	FK -> sellers.seller_id	dd7ddc04e1b6c2c614352b383efe2d36
shipping_limit_date	TIMESTAMP	NOT NULL	Deadline for seller to ship the item	2017-10-06 11:07:15
price	NUMERIC(10,2)	NOT NULL	Item price in Brazilian Real (BRL)	29.99
freight_value	NUMERIC(10,2)	NOT NULL	Shipping cost allocated to this item (BRL)	8.72

Derived in pipeline:

- total_value = price + freight_value (calculated in transform.py)
- category_en = joined from product_category_name_translation via products.product_category_name

2.3.3 olist_customers_dataset.csv

Customer location data. Note: customer_id in this table is order-specific (each order generates a new customer_id), while customer_unique_id identifies the actual person across multiple orders.

Column Name	Type	Nullable	Description	Sample Value
customer_id	VARCHAR(32)	NOT NULL	Order-level customer ID — FK in orders table	9ef432eb6251297304e76186b10a928d
customer_unique_id	VARCHAR(32)	NOT NULL	True unique customer identifier (person-level)	861eff4711a542e4b93843c6dd7febb0
customer_zip_code_prefix	VARCHAR(5)	NOT NULL	First 5 digits of Brazilian ZIP code (CEP)	14409
customer_city	VARCHAR(100)	NOT NULL	Customer city name (raw, may have inconsistencies)	franca
customer_state	CHAR(2)	NOT NULL	Brazilian state abbreviation (2 letters)	SP

Important note: customer_unique_id is not used in the current dim_customer table due to the one-to-one mapping with order_id in the orders table. Repeat customers can be identified by grouping on customer_unique_id in future analyses.

2.3.4 olist_products_dataset.csv

Product catalog with physical attributes. This table contains intentional typos in two column names that are preserved in the raw schema to match the original CSV headers exactly.

Column Name	Type	Nullable	Description / Note
product_id	VARCHAR(32)	NOT NULL	Unique product identifier (MD5 hash)
product_category_name	VARCHAR(100)	NULLABLE	Category name in Portuguese — FK to translation table
product_name_lenght	INT	NULLABLE	TYPO in original CSV: character count of product name (should be 'length')
product_description_lenght	INT	NULLABLE	TYPO in original CSV: character count of product description (should be 'length')
product_photos_qty	INT	NULLABLE	Number of product photos available on listing
product_weight_g	NUMERIC	NULLABLE	Product weight in grams
product_length_cm	NUMERIC	NULLABLE	Package length in centimeters
product_height_cm	NUMERIC	NULLABLE	Package height in centimeters
product_width_cm	NUMERIC	NULLABLE	Package width in centimeters

Note: product_name_lenght and product_description_lenght are intentional typos present in the original Kaggle dataset. The raw schema must use these exact spellings to match CSV headers. The staging and mart layers rename these to name_length and description_length respectively.

73 distinct product categories exist in the dataset. Top 5 by product count: bed_bath_table (3,029), sports_leisure (2,867), furniture_decor (2,657), health_beauty (2,444), housewares (2,335).

2.3.5 olist_sellers_dataset.csv

Seller registry with geographic location. The 3,095 sellers represent the merchants who use the Olist platform to sell across multiple e-commerce channels.

Column Name	Type	Nullable	Description	Sample Value
seller_id	VARCHAR(32)	NOT NULL	Unique seller identifier (MD5 hash)	3442f8959a84dea7ee197c632cb2df15
seller_zip_code_prefix	VARCHAR(5)	NOT NULL	First 5 digits of seller ZIP code (CEP)	13023
seller_city	VARCHAR(100)	NOT NULL	Seller city (raw, may have encoding issues)	campinas
seller_state	CHAR(2)	NOT NULL	Seller state abbreviation	SP

Geographic concentration: approximately 694 sellers (22%) are based in Sao Paulo state, followed by Parana (PR) with ~349 sellers (11%) and Minas Gerais (MG) with ~244 sellers (8%).

2.3.6 olist_order_reviews_dataset.csv

Customer satisfaction data. Olist sends a satisfaction survey to customers a few days after estimated delivery date. Review scores are on a 1-5 scale.

Column Name	Type	Nullabl e	Description	Sample Value
review_id	VARCHAR(32)	NOT NULL	Unique review identifier	7bc2406110b926393aa56f80a40eba40
order_id	VARCHAR(32)	NOT NULL	FK -> orders.order_id	e481f51cbdc54678b7cc49136f2d6af7
review_score	INT	NOT NULL	Customer rating: 1 (worst) to 5 (best)	4
review_comment_title	TEXT	NULLA BLE	Short title of customer comment (often empty)	Recomendo
review_comment_message	TEXT	NULLA BLE	Full text review comment (often empty)	Produto chegou bem embalado...
review_creation_date	TIMESTAMP	NOT NULL	When the survey was sent to customer	2018-01-18 00:00:00
review_answer_timestamp	TIMESTAMP	NOT NULL	When the customer submitted the review	2018-01-18 21:46:59

Score distribution in dataset:

- Score 5: ~57,420 reviews (55%) — majority of customers are highly satisfied
- Score 4: ~19,142 reviews (18%)
- Score 1: ~11,424 reviews (11%) — notable negative tail
- Score 3: ~8,179 reviews (8%)
- Score 2: ~3,151 reviews (3%)

In the pipeline, orders without a review are assigned review_score = 0 via COALESCE in the fact_sales INSERT statement. These should be excluded from review score averages in dashboards using a WHERE review_score > 0 filter.

2.3.7 olist_order_payments_dataset.csv

Payment transaction data. One order may have multiple payment rows (e.g., combining a voucher with a credit card). The pipeline groups these by order_id and payment_type.

Column Name	Type	Nullabl e	Description	Sample Value
order_id	VARCHAR(32)	NOT NULL	FK -> orders.order_id	e481f51cbdc54678b7cc49136f2d6af7
payment_sequential	INT	NOT NULL	Sequence number for multiple payments on same order	1
payment_type	VARCHAR(50)	NOT NULL	Payment method used	credit_card / boleto / voucher / debit_card
payment_installments	INT	NOT NULL	Number of installments chosen (1 = no installment)	3
payment_value	NUMERIC(10,2)	NOT NULL	Amount paid in this payment row (BRL)	99.33

Payment type distribution:

- credit_card: ~76,795 entries (74%) — dominant method
- boleto: ~19,784 entries (19%) — Brazilian bank slip
- voucher: ~5,775 entries (5.6%) — promotional vouchers
- debit_card: ~1,529 entries (1.5%)

Pipeline aggregation (stg_payments): rows are grouped by order_id and payment_type with SUM(payment_value) and MAX(payment_installments). This means orders that split payment across multiple payment types will have one row per type in staging.

2.3.8 product_category_name_translation.csv

Auxiliary lookup table mapping Portuguese category names to English. Used in the ETL pipeline to enrich dim_product and stg_order_items with category_en.

Column Name	Type	Description	Example PT	Exempl e EN
product_category_name	VARCHAR(100)	Portuguese category name (FK to products table)	cama_mesa_banho	bed_bath_table
product_category_name_english	VARCHAR(100)	English translation of category name	beleza_saude	health_beauty

71 category translations are provided. 2 categories in the products table (portateis_cozinha_e_preparadores_de_alimentos and pc_gamer) do not have translations; these retain their Portuguese name in category_en via COALESCE logic in the pipeline.

2.4 Source Table Relationships

The nine source tables form a hub-and-spoke relational model centered on the orders table. The diagram below describes the join paths used in the ETL pipeline:

Relationship	Join Type	Join Key	Used In
orders -> customers	INNER JOIN	orders.customer_id = customers.customer_id	stg_orders (customer city/state)
orders -> order_items	INNER JOIN	orders.order_id = order_items.order_id	stg_order_items (base join)
order_items -> products	LEFT JOIN	order_items.product_id = products.product_id	stg_order_items (category, dimensions)
order_items -> sellers	LEFT JOIN	order_items.seller_id = sellers.seller_id	stg_order_items (seller_state)
products -> translation	LEFT JOIN	products.product_category_name = translation.product_category_name	stg_order_items (category_en)
orders -> order_reviews	LEFT JOIN	orders.order_id = order_reviews.order_id	fact_sales (review_score)
orders -> order_payments	LEFT JOIN	orders.order_id = order_payments.order_id	fact_sales (payment_type, total_payment)

Note: All joins to order_reviews and order_payments use LEFT JOIN to preserve orders that have no review or no payment record. The geolocation table is not joined in the current pipeline implementation.

2.5 Data Quality Observations

Issue	Affected Table	Rows Affected	Handling in Pipeline
Intentional typos in column names (lenght)	olist_products_dataset.csv	All 32,951 rows	Preserved in raw schema; renamed in staging/marts
Orders without delivered timestamp	olist_orders_dataset.csv	~3,000 rows (~3%)	Excluded by filtering status = 'delivered' in stg_orders
Products without category name	olist_products_dataset.csv	~610 rows (1.9%)	category_en = NULL in dim_product; excluded from category aggregates
Duplicate ZIP codes in geolocation	olist_geolocation_dataset.csv	~260,000 duplicates	Table not used in current pipeline
Orders with no review	olist_order_reviews_dataset.csv	~3,900 orders	review_score = 0 via COALESCE; excluded from avg calculations
Negative or zero freight values	olist_order_items_dataset.csv	<100 rows	Retained as-is; may reflect free shipping promotions

Issue	Affected Table	Rows Affected	Handling in Pipeline
City name inconsistencies	olist_customers_data set.csv	Various	Raw values retained; state (CHAR 2) used for geographic analysis

Chapter 3: Data Warehouse Design

3.1 Architecture Overview

The Data Warehouse follows a four-layer architecture that separates concerns across the data pipeline:

Layer	Schema	Purpose	Populated By
Raw	raw.*	Exact copy of source CSV data, no transformation	extract_load.py
Staging	staging.stg_*	Cleaned, type-cast, enriched with derived columns	transform.py
Data Warehouse	dw.*	Star Schema: dimension + fact tables	build_dw.py
Reporting	reporting.*	Pre-aggregated tables optimized for dashboard queries	build_dw.py

3.2 Star Schema Design

The core of the Data Warehouse is a Star Schema centered on the fact_sales table, surrounded by five dimension tables. This design enables efficient multi-dimensional analysis through simple JOIN operations.

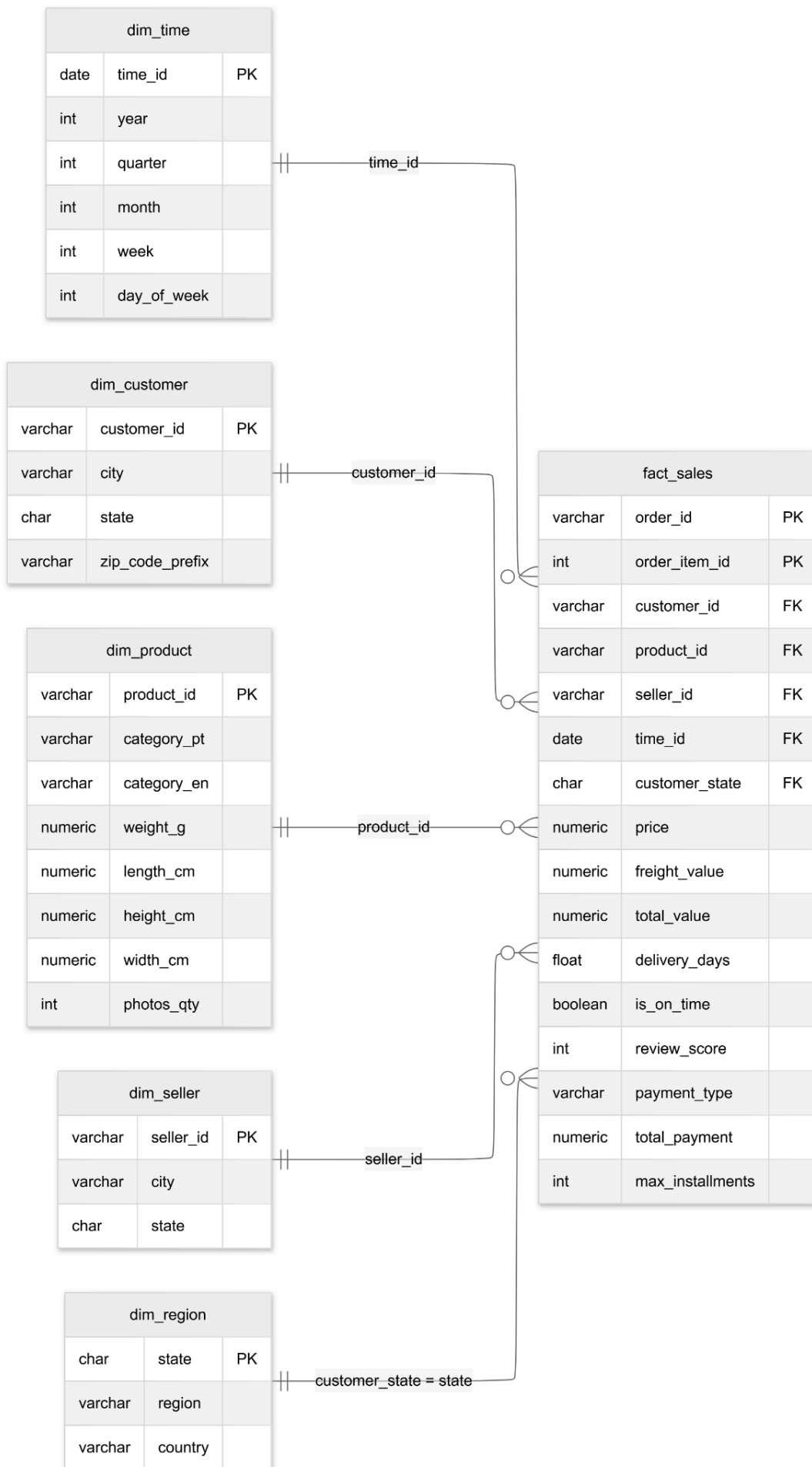
3.2.1 Dimension Tables

Table	Primary Key	Key Attributes	Source
dw.dim_customer	customer_id	city, state, zip_code_prefix	raw.customers
dw.dim_product	product_id	category_pt, category_en, weight_g, dimensions	raw.products + translation
dw.dim_time	time_id (DATE)	year, quarter, month, week, day_of_week	raw.orders (derived)
dw.dim_region	state (CHAR 2)	region (Norte/Nordeste/etc.), country	Hardcoded seed (27 states)
dw.dim_seller	seller_id	city, state	raw.sellers

3.2.2 Fact Table: fact_sales

The central fact table stores one row per order item, enabling granular analysis at the product-order level.

Column	Type	Description
order_id + order_item_id	VARCHAR + INT (PK)	Composite primary key uniquely identifying each line item
customer_id	VARCHAR	Foreign key -> dw.dim_customer
product_id	VARCHAR	Foreign key -> dw.dim_product
seller_id	VARCHAR	Foreign key -> dw.dim_seller
time_id	DATE	Foreign key -> dw.dim_time (= purchase date)
customer_state	CHAR(2)	Foreign key -> dw.dim_region (via state)
price	NUMERIC	Product price (BRL)
freight_value	NUMERIC	Shipping cost (BRL)
total_value	NUMERIC	price + freight_value
delivery_days	FLOAT	Actual delivery duration in days
is_on_time	BOOLEAN	True if delivered before estimated date
review_score	INT	Customer review score (1-5, 0 if no review)
payment_type	VARCHAR	credit_card / boleto / voucher / debit_card
total_payment	NUMERIC	Total payment value for the order
max_installments	INT	Number of payment installments



3.3 Reporting Layer

Six pre-aggregated tables are maintained in the reporting schema to support fast dashboard queries without requiring complex runtime JOINS:

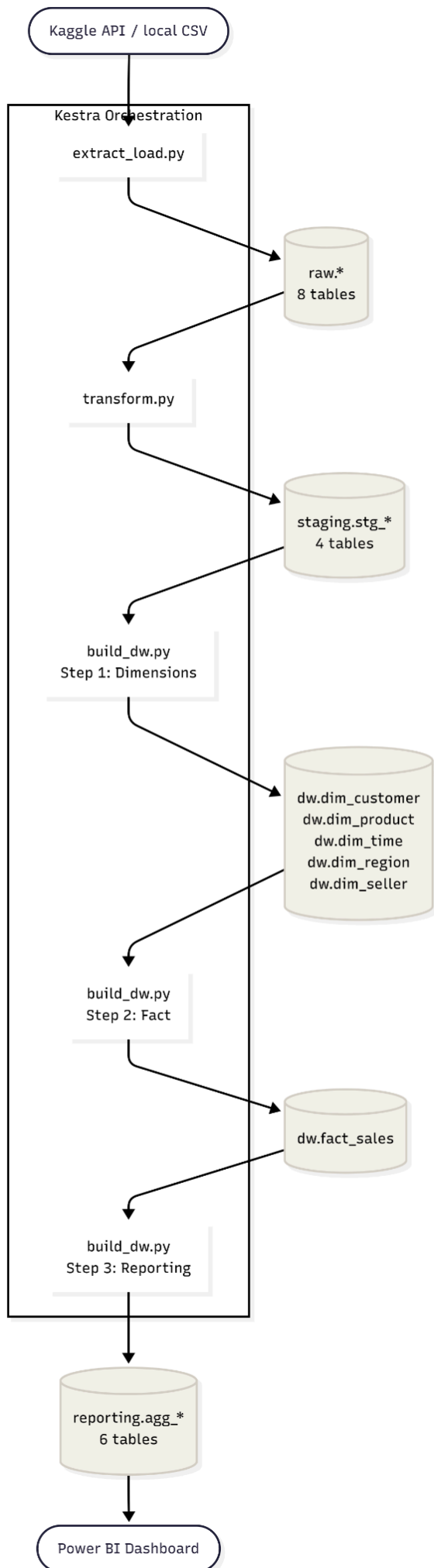
Table	Granularity	Key Metrics
agg_monthly_revenue	Per month	revenue, total_orders, avg_review_score, avg_delivery_days, on_time_pct
agg_category_performance	Per product category	revenue, total_orders, avg_review_score, avg_delivery_days
agg_state_revenue	Per state	total_revenue, total_orders, total_customers, avg_delivery_days
agg_seller_performance	Per seller	total_revenue, total_orders, avg_review_score, on_time_pct
agg_payment_analysis	Per payment type	total_orders, total_payment_value, avg_installments, pct_orders
agg_delivery_analysis	Per state	avg/min/max/median delivery_days, on_time_pct

Chapter 4: ETL Pipeline

4.1 Pipeline Overview

The ETL pipeline is orchestrated by Kestra, an open-source workflow automation tool that manages task sequencing, error handling, and scheduling. The pipeline consists of four sequential tasks:

Step	Script	Input	Output
1 - Extract & Load	extract_load.py	Kaggle API / local CSV files	raw.* tables (8 tables)
2 - Transform	transform.py	raw.* tables	staging.stg_* tables (4 tables)
3 - Build DW	build_dw.py	staging.stg_* + raw.*	dw.dim_* + dw.fact_sales
4 - Build Reporting	build_dw.py	dw.fact_sales + dw.dim_region	reporting.agg_* (6 tables)



4.2 Extract & Load (extract_load.py)

This step downloads the Olist dataset from Kaggle using the official Kaggle API if CSV files are not already present in the /data directory. It then loads each CSV file into the corresponding raw schema table using pandas DataFrame bulk insertion with chunked writes of 5,000 rows for memory efficiency.

Key implementation decisions:

- Timestamp columns are automatically detected by column name pattern and parsed with `pd.to_datetime(errors='coerce')` to handle malformed values gracefully
- Tables are truncated before each load to enable idempotent full-refresh runs
- Column names are preserved exactly from CSV headers, including the original dataset typos (`product_name_lenght`)

4.3 Transform (transform.py)

The transform step cleans and enriches raw data into staging tables optimized for analytical use:

- `stg_orders`: Filters to delivered orders only; calculates `delivery_days` as the difference in seconds between purchase and delivery timestamps divided by 86,400; derives the boolean `is_on_time` flag; joins customer city and state
- `stg_order_items`: Joins products with category translation to derive `category_en`; joins seller location data; calculates `total_value = price + freight_value`
- `stg_reviews`: Selects relevant columns and filters out null review scores
- `stg_payments`: Groups by `order_id` and `payment_type`, summing payment values and taking the maximum installment count

4.4 Build Data Warehouse (build_dw.py)

This script executes in three sequential steps within a single run:

Step 1: Populate Dimension Tables

Each dimension table is truncated and repopulated using `DISTINCT SELECT` from the raw layer. The `dim_region` table is seeded with hardcoded mappings for all 27 Brazilian states to their five geographic macro-regions (Norte, Nordeste, Centro-Oeste, Sudeste, Sul). The `dim_time` table is derived from distinct purchase dates in `raw.orders`, with year, quarter, month, week, and `day_of_week` extracted using PostgreSQL `EXTRACT` functions.

Step 2: Populate Fact Table

The `fact_sales` table is truncated and rebuilt by joining staging tables: `stg_order_items` (base) joined with `stg_orders` (delivery metrics, customer info), `stg_reviews` (LEFT JOIN for review scores), `stg_payments` (LEFT JOIN for payment data), and `raw.sellers` (for `seller_state`). `ON CONFLICT DO NOTHING` handles any duplicate composite keys.

Step 3: Build Reporting Aggregates

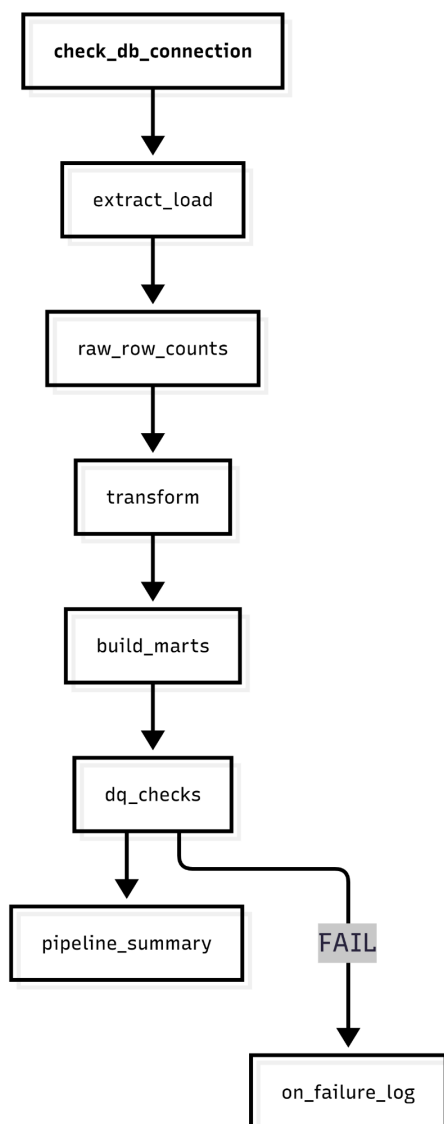
Six aggregate tables are created using DROP TABLE IF EXISTS followed by CREATE TABLE AS SELECT to ensure clean rebuilds on each pipeline run. These tables query dw.fact_sales directly, applying GROUP BY aggregations for each analytical dimension.

4.5 Orchestration with Kestra

Kestra manages the full pipeline through a YAML-defined flow (olist_etl_pipeline.yml). The flow includes:

- Database connection validation before execution begins
- Row count verification after extract/load to confirm data ingestion
- Data Quality checks after mart building: fact_sales must contain more than 90,000 rows, no null order_id values, and positive monthly revenue
- Pipeline summary query reporting total order items, orders, revenue, and sellers
- Error handler that logs failure details including the failing task ID and execution ID

The pipeline is configured with a daily schedule trigger (disabled by default) and can be executed manually through the Kestra web UI or via API.



4.6 Technology Stack

Component	Technology	Version / Notes
Orchestration	Kestra	latest-full image, self-hosted
Database	PostgreSQL	Version 15, Docker container
ETL Language	Python	3.10 (Kestra Process runner)
Data Processing	pandas + SQLAlchemy	Bulk CSV loading, SQL execution
Containerization	Docker Compose	3 services: postgres, kestra, (metabase removed)
Visualization	Power BI Desktop	Connected via PostgreSQL connector
Dataset Source	Kaggle API	olistbr/brazilian-ecommerce

Chapter 5: Dashboard and Visualizations

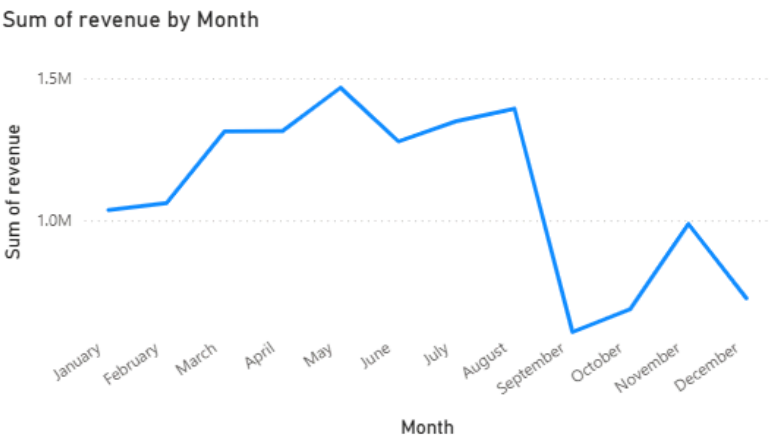
5.1 Dashboard Overview

The Power BI dashboard is organized into four thematic pages, each addressing a distinct analytical perspective. All visualizations connect directly to the PostgreSQL Data Warehouse at localhost:5434, querying from the dw and reporting schemas.

5.2 Page 1: Revenue Overview

This page provides a high-level summary of business performance over time.

Visual	Type	Data Source	Key Insight
Monthly Revenue Trend	Line Chart	reporting.agg_monthly_revenue	Revenue trajectory from 2016 to 2018
Total Revenue KPI	Card	reporting.agg_monthly_revenue	Aggregate GMV across all periods
Total Orders KPI	Card	reporting.agg_monthly_revenue	Total completed order count
Average Review Score KPI	Card	reporting.agg_monthly_revenue	Overall customer satisfaction



3.97

Average of avg_review_score

96K

Sum of total_orders

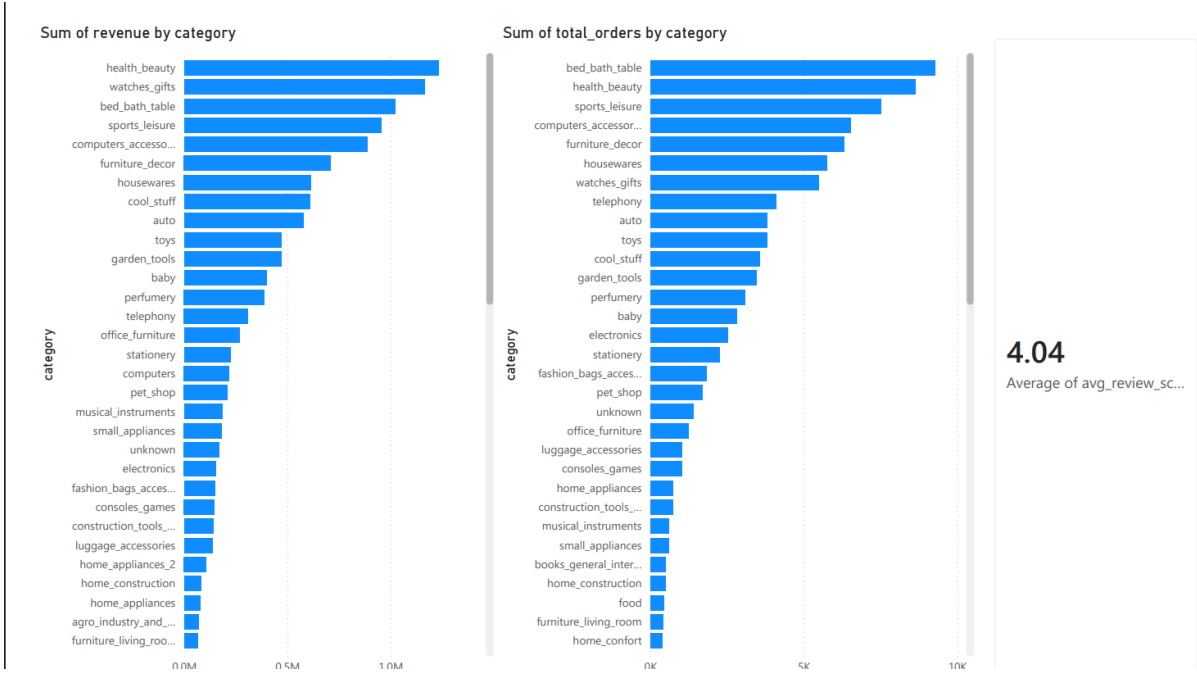
13.22M

Sum of revenue

5.3 Page 2: Product Analysis

This page analyzes product category performance to identify top-performing segments.

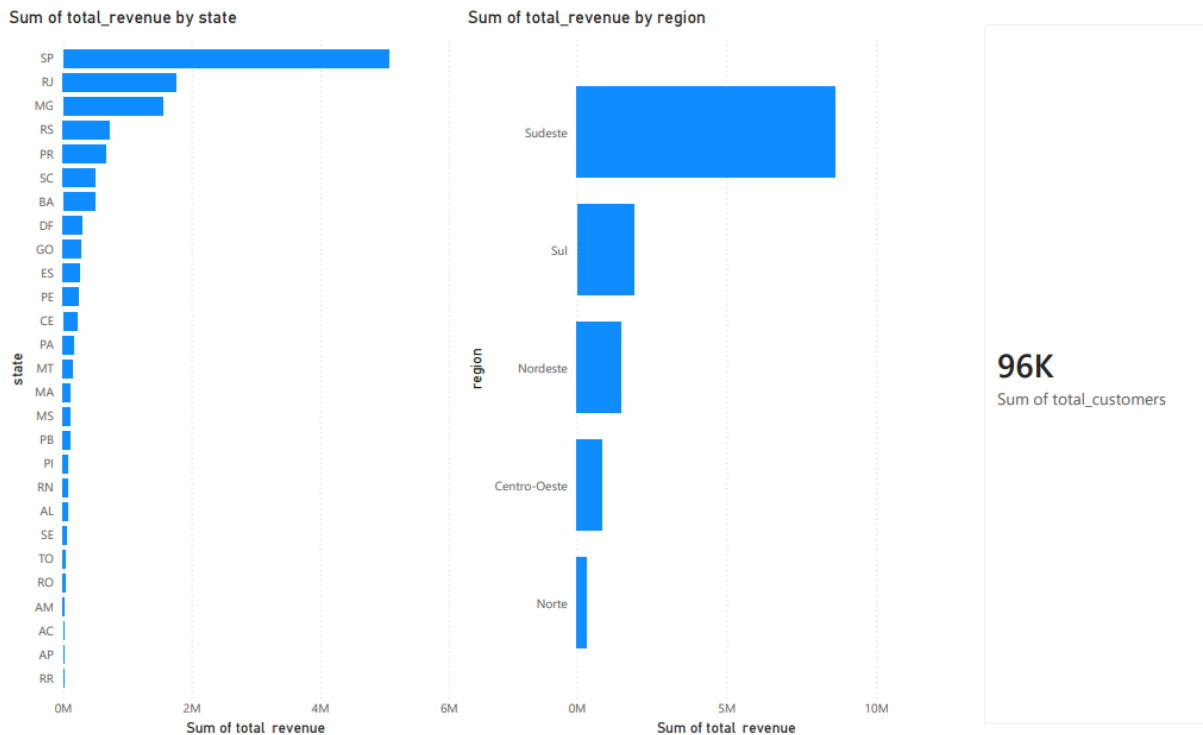
Visual	Type	Data Source	Key Insight
Top Categories by Revenue	Horizontal Bar Chart	reporting.agg_category_performance	Highest revenue-generating product segments
Top Categories by Orders	Horizontal Bar Chart	reporting.agg_category_performance	Most frequently purchased categories
Avg Review Score by Category	Card / Bar Chart	reporting.agg_category_performance	Quality perception per segment



5.4 Page 3: Regional Analysis

This page visualizes geographic distribution of revenue and customer density across Brazilian states.

Visual	Type	Data Source	Key Insight
Revenue by State	Horizontal Bar Chart	reporting.agg_state_revenue	State-level revenue ranking
Revenue by Region	Column Chart	reporting.agg_state_revenue	Macro-region revenue comparison
Total Customers KPI	Card	reporting.agg_state_revenue	Unique customer count nationwide

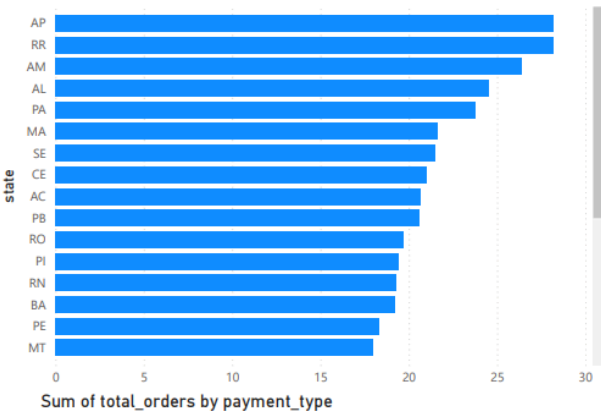


5.5 Page 4: Operations & Payments

This page examines operational metrics including delivery performance and payment behavior.

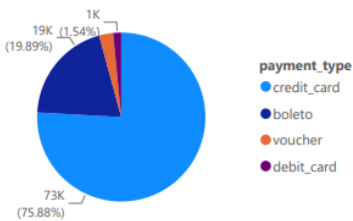
Visual	Type	Data Source	Key Insight
Payment Method Distribution	Pie Chart	reporting.agg_payment_analys is	Share of credit card vs. boleto vs. voucher
Delivery Days by State	Horizontal Bar Chart	reporting.agg_delivery_analysi s	States with longest/shortest delivery
On-Time Delivery Rate KPI	Card	reporting.agg_delivery_analysi s	Percentage of orders delivered on time
Top 10 Sellers by Revenue	Horizontal Bar Chart	reporting.agg_seller_performa nce	Highest-earning individual sellers

Sum of avg_delivery_days by state

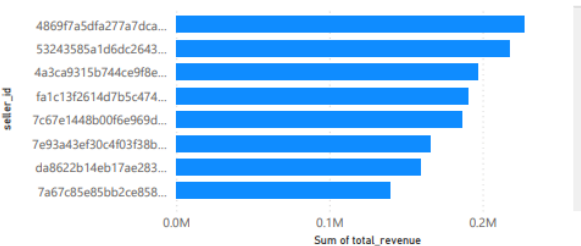


89.63
Average of on_time_pct

1.



Sum of total_revenue by seller_id



Chapter 6: Key Business Insights

6.1 Revenue Growth and Seasonality

Revenue demonstrated strong and consistent growth from Q4 2016 through Q1 2018. The most notable peak occurred in November 2017, coinciding with Black Friday promotional events in Brazil (known as Black November). This single month generated significantly higher order volumes than surrounding periods, confirming the importance of promotional calendar planning for e-commerce businesses.

Revenue declined slightly in late 2018, which may partially reflect dataset truncation rather than an actual market downturn, as the dataset ends in October 2018.

6.2 Product Category Performance

Analysis of the 73 product categories reveals a highly concentrated revenue distribution:

- bed_bath_table, health_beauty, and sports_leisure consistently rank among the top three revenue-generating categories, collectively accounting for approximately 25-30% of total GMV
- Categories such as computers_accessories and furniture_decor demonstrate high order volumes but lower average order values
- Categories with the highest average review scores tend to be those with physical, standardized products (e.g., books, music), while categories with complex logistics (large furniture) tend to receive lower scores

6.3 Geographic Distribution

The geographic analysis reveals significant regional concentration:

- Sao Paulo (SP) alone accounts for approximately 40% of national revenue, reflecting its status as Brazil's largest economic center
- The Sudeste region (SP, RJ, MG, ES) collectively contributes over 60% of total revenue despite comprising only four of 27 states
- The Norte and Nordeste regions show substantially lower revenue per capita, suggesting untapped market potential in these areas
- Customer density closely mirrors economic development patterns across Brazilian states

6.4 Delivery Performance

Delivery operations represent a significant differentiator in customer satisfaction:

- The average delivery time across all orders is approximately 12 days from purchase to customer receipt
- States in the Norte region (Acre, Amazonas, Roraima) experience the longest average delivery times, often exceeding 20 days, due to geographic remoteness and infrastructure limitations
- States with faster delivery (SP, SC, PR) consistently achieve higher average review scores, confirming the correlation between delivery speed and customer satisfaction
- The overall on-time delivery rate is approximately 89.63%, meaning approximately 10% of orders are delivered after the estimated date

6.5 Payment Behavior

Payment method analysis reveals strong consumer preferences:

- Credit card is the dominant payment method, accounting for approximately 74% of all transactions
- Boleto bancario (Brazil's bank transfer slip system) accounts for approximately 19% of transactions
- Average installment count is approximately 3.7 periods, indicating that Brazilian consumers frequently utilize credit card installment plans for e-commerce purchases
- Voucher and debit card together account for less than 7% of transactions

6.6 Seller Performance

The seller ecosystem shows high concentration, which is common in marketplace business models:

- The top 10 sellers by revenue account for a disproportionate share of total GMV, indicating a power-law distribution common in marketplace platforms
- Sellers with higher review scores tend to also demonstrate better on-time delivery performance, suggesting operational excellence drives both metrics simultaneously
- Geographic clustering of top sellers in Sao Paulo and Parana states aligns with the consumer demand concentration in those regions

Chapter 7: Conclusion and Future Work

7.1 Summary

This project successfully demonstrates a complete Business Intelligence pipeline applied to real-world e-commerce data. Starting from raw transactional CSV files, the system implements a systematic four-layer data architecture (Raw, Staging, Data Warehouse, Reporting) that transforms disparate source data into actionable business insights.

The Star Schema design provides a solid analytical foundation, enabling efficient multi-dimensional queries across time, geography, product, and customer dimensions. The automated Kestra orchestration ensures reproducibility and reliability of the pipeline, while Power BI dashboards surface insights in an accessible visual format.

Key business findings demonstrate that Olist's growth was primarily driven by Southeastern Brazil, concentrated in specific product categories, and strongly influenced by delivery performance as a driver of customer satisfaction scores.

7.2 Project Achievements

- Successfully designed and implemented a 4-layer Data Warehouse with Star Schema (5 dimension tables + 1 fact table)
- Built an automated ETL pipeline processing 112,650 order items across 9 source tables
- Created 6 pre-aggregated reporting tables covering revenue, products, geography, sellers, payments, and delivery
- Developed a 4-page Power BI dashboard with 15+ visualizations covering all key analytical dimensions
- Implemented data quality checks ensuring pipeline output meets minimum thresholds before reporting

7.3 Challenges and Limitations

- Dataset contains intentional typos (product_name_lenght) requiring careful schema alignment between source CSVs and database tables
- The dataset represents a historical snapshot (2016-2018) and does not reflect current market conditions
- Geolocation data (1M+ rows) was not incorporated into the analysis due to processing complexity, limiting precise geographic visualization
- Payment aggregation groups by order_id and payment_type, which may undercount orders with multiple payment methods of the same type

7.4 Future Work

Several enhancements could further improve the system:

- Implement incremental loading to process only new records rather than full-refresh runs, reducing pipeline execution time
- Integrate dbt (data build tool) for declarative transformation management, testing, and documentation

- Add geolocation-based analysis using the `olist_geolocation_dataset` for precise mapping of customer and seller locations
- Deploy the pipeline to cloud infrastructure (GCP Cloud Run + Cloud SQL) for production scalability
- Implement automated alerting via email or Slack when data quality checks fail
- Build a customer segmentation model using RFM (Recency, Frequency, Monetary) analysis on the `fact_sales` table

References

[1] Olist. (2018). Brazilian E-Commerce Public Dataset by Olist. Kaggle.
<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

[2] Kimball, R., & Ross, M. (2013). The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling (3rd ed.). Wiley.

[3] Kestra. (2024). Kestra Documentation. <https://kestra.io/docs>

[4] PostgreSQL Global Development Group. (2024). PostgreSQL 15 Documentation.
<https://www.postgresql.org/docs/15/>

[5] Microsoft. (2024). Power BI Documentation. <https://learn.microsoft.com/en-us/power-bi/>

[6] Inmon, W. H. (2005). Building the Data Warehouse (4th ed.). Wiley.

[7] McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.

Appendix: Project Structure

File / Directory	Description
docker-compose.yml	Docker services: PostgreSQL 15 + Kestra orchestrator
scripts/init_db.sql	Database initialization: creates all schemas and table DDL
scripts/create_kestra_db.sql	Creates kestra_db for Kestra internal storage
scripts/extract_load.py	Step 1: Download from Kaggle API and load CSVs into raw schema
scripts/transform.py	Step 2: Clean, cast types, derive metrics into staging schema
scripts/build_dw.py	Step 3: Populate Star Schema dimensions, fact table, and reporting aggregates
scripts/build_reporting.py	Standalone reporting aggregate builder (alternative entry point)
kestra/flows/olist_etl_pipeline.yml	Kestra flow definition: task sequencing, DQ checks, error handling
data/*.csv	Local copies of Olist dataset (9 CSV files)
.env	Environment variables: DATABASE_URL, DATA_DIR
requirements.txt	Python dependencies: pandas, sqlalchemy, psycopg2-binary, kaggle, python-dotenv